Edgeflow_

A Real-Time, Low-Latency Market Microstructure Engine for Cryptocurrency Spot Markets

Composite Alpha Signal Construction, Multi-Horizon Predictive Validation,
and Realistic Execution Simulation in C++20

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March 2025

JEL Classification: G12, G14, G15, C63, C55 | Keywords: High-Frequency Trading, Market Microstructure, Order Book Imbalance, Alpha Decay, Execution Simulation, C++, Binance, Cryptocurrency

Abstract

We present edgeflow, a production-grade, real-time market microstructure engine for Binance spot markets, implemented in C++20. The system streams live Level-2 order book data and aggregated trade flow via WebSocket, constructs a composite alpha signal from four microstructure features, validates its predictive power across four time horizons (50ms, 100ms, 250ms, 500ms) without lookahead bias, and simulates round-trip trade execution with realistic fee modelling and inventory constraints.

The composite signal integrates Order Book Imbalance at L1 (weight 0.30), Order Book Imbalance at L5 (weight 0.20), microprice offset from mid (weight 0.20), and trade flow imbalance over a rolling window (weight 0.30). Each depth update is classified into a spread regime - TIGHT, NORMAL, or WIDE - and the signal's predictive performance is reported by bucket and regime to identify conditions under which microstructure signals carry statistically significant alpha.

The architecture implements Binance's diff-depth synchronization protocol with full sequence-ID validation and gap detection, ensuring the maintained L2 book is always consistent with the exchange's canonical state. The execution simulator supports both aggressive (taker) and passive (maker) entry modes, per-trade fee deduction, passive fill timeouts, and maximum inventory limits. The validator reports hit rate, average price move in basis points, and Sharpe ratio by signal bucket and horizon, along with an alpha decay curve quantifying the half-life of the microstructure signal. Preliminary live-data results indicate that the composite signal achieves hit rates above 60% at the 50ms horizon in the TIGHT spread regime for BTCUSDT, with a raw Sharpe exceeding 2.0 in simulation.

1. Introduction

Market microstructure - the study of how latent supply and demand are translated into observable prices and quantities through the mechanics of trading - has emerged as a central discipline in quantitative finance since the foundational work of Kyle (1985) and Glosten & Milgrom (1985). The proliferation of electronic limit order books, the decimalization of tick sizes, and the rise of algorithmic and high-frequency trading (HFT) have collectively elevated the practical importance of microstructure theory: the ability to extract short-horizon predictive signals from the live order book is now a primary source of edge for firms competing on sub-millisecond timescales.

Cryptocurrency spot markets, and Binance in particular, present a structurally interesting laboratory for microstructure research. Unlike regulated equity exchanges, Binance operates continuously (24/7/365), offers a public REST and WebSocket API with millisecond-resolution depth data, and exhibits a broad range of liquidity regimes across its listed pairs. The market is populated by a diverse mix of retail participants, proprietary trading firms, arbitrageurs, and market makers, creating rich cross-sectional variation in microstructure dynamics.

The primary challenge in live microstructure research is systems engineering: reliable signal extraction at sub-second timescales requires correct implementation of exchange protocols, zero-copy data pipelines, and lock-free data structures - requirements that Python or interpreted languages cannot efficiently satisfy. C++ remains the lingua franca of production HFT systems precisely because it provides deterministic memory management, compiler-level optimization, and direct access to OS-level timing and network primitives.

edgeflow is a complete, open-source implementation of a microstructure engine addressing all of these requirements. Its four-component architecture - OrderBook, SignalEngine, MultiHorizonValidator, ExecutionSimulator - maps cleanly onto the four stages of the HFT alpha pipeline: data acquisition, signal construction, signal validation, and strategy simulation. This paper provides a rigorous academic treatment of each component, situating the implementation choices within the existing microstructure literature and documenting the empirical properties of the resulting signals.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the system architecture and each component in detail. Section 4 formalizes the signal construction methodology. Section 5 describes the multi-horizon validation framework. Section 6 documents the execution simulation model. Section 7 presents empirical results. Section 8 discusses limitations and extensions. Section 9 concludes.

2. Literature Review

2.1 Theoretical Foundations of Order Book Microstructure

The canonical decomposition of the bid-ask spread into adverse selection, inventory, and order-processing components (Glosten & Milgrom, 1985; Hasbrouck, 1988) establishes the theoretical basis for order book-based alpha signals. Informed traders possess private information about future price direction; their order flow

imbalance - buying more aggressively when they expect prices to rise - is detectable in the limit order book in the moments before the information is impounded into the mid price. This adverse selection component is the theoretical underpinning of the Order Book Imbalance (OBI) signal.

Glosten & Harris (1988) decompose the price impact of trades into a permanent component (information) and a transient component (liquidity provision costs). The permanent component corresponds to the OBI signal's predictive horizon; the transient component corresponds to the bid-ask spread that must be overcome in execution. Madhavan, Richardson & Roomans (1997) extend this to a dynamic model, estimating the fraction of price movement attributable to public versus private information.

2.2 Order Book Imbalance as a Predictive Signal

Cont, Kukanov & Stoikov (2014) provide the first rigorous statistical analysis of order book imbalance as a short-horizon price predictor. Using high-frequency equity data, they show that OBI - defined as the imbalance between best-bid and best-ask queue sizes - is a statistically significant predictor of mid-price changes at horizons from 1 to 10 events. Crucially, they show that OBI is informative over and above trade direction, establishing it as an independent signal source.

Stoikov (2018) extends OBI to the microprice - a quantity-weighted average of the best bid and ask that provides a better estimate of the true efficient price than the arithmetic mid. The microprice's deviation from mid is the third signal component in edgeflow and is theoretically motivated as a measure of the imbalance between liquidity supply and demand at the inside quotes.

Sirignano & Cont (2019) demonstrate that deep neural networks trained on the full L2 order book can predict short-horizon price movements with higher accuracy than OBI-based linear models, but require substantial training data and are computationally expensive to deploy in real time. The linear composite signal in edgeflow represents the computationally tractable benchmark against which such nonlinear models can be evaluated.

2.3 Trade Flow Imbalance

Chordia & Subrahmanyam (2004) document that trade imbalance - the excess of buyer-initiated over seller-initiated trades — predicts short-horizon returns in equity markets. Lee & Ready (1991) develop the standard algorithm for classifying trade direction from transaction data; in exchange-provided aggTrade data (as used in edgeflow), direction is provided directly by the exchange, eliminating classification error. Easley, Kiefer & O'Hara (1997) formalize the PIN (Probability of Informed Trading) model, which uses trade arrival rates to infer the presence of informed traders - a structural motivation for including trade flow as a signal component.

2.4 Alpha Decay in High-Frequency Signals

The half-life of microstructure signals is a central empirical question in HFT research. Bouchaud et al. (2018) document that the price impact of individual trades decays on a timescale of approximately 10–100 seconds in equity markets, with the decay curve well-approximated by a power law. Almgren & Chriss (2001) provide the theoretical framework for optimal execution in the presence of transient price impact, which is the benchmark

for the execution simulation in edgeflow. The alpha decay curve reported by the MultiHorizonValidator directly quantifies the useful predictive horizon of the composite signal.

2.5 Cryptocurrency Microstructure

Makarov & Schoar (2020) document substantial price discrepancies across cryptocurrency exchanges, attributable to capital flow constraints - establishing that crypto markets are not fully integrated and that microstructure-level price discovery differs meaningfully from equity markets. Cong, Li & Wang (2021) provide evidence of informed trading in Bitcoin markets, consistent with the adverse selection model. Brauneis & Mestel (2018) document that Bitcoin exhibits lower market efficiency than equities at short horizons, suggesting that microstructure signals may carry higher predictive power in crypto than in mature equity markets.

3. System Architecture

3.1 Overview

edgeflow is structured as a four-stage pipeline in which data flows from the exchange feed through book maintenance, signal computation, and validation/simulation in a single-threaded processing loop, with a dedicated WebSocket thread handling network I/O asynchronously. The architecture is illustrated schematically below.

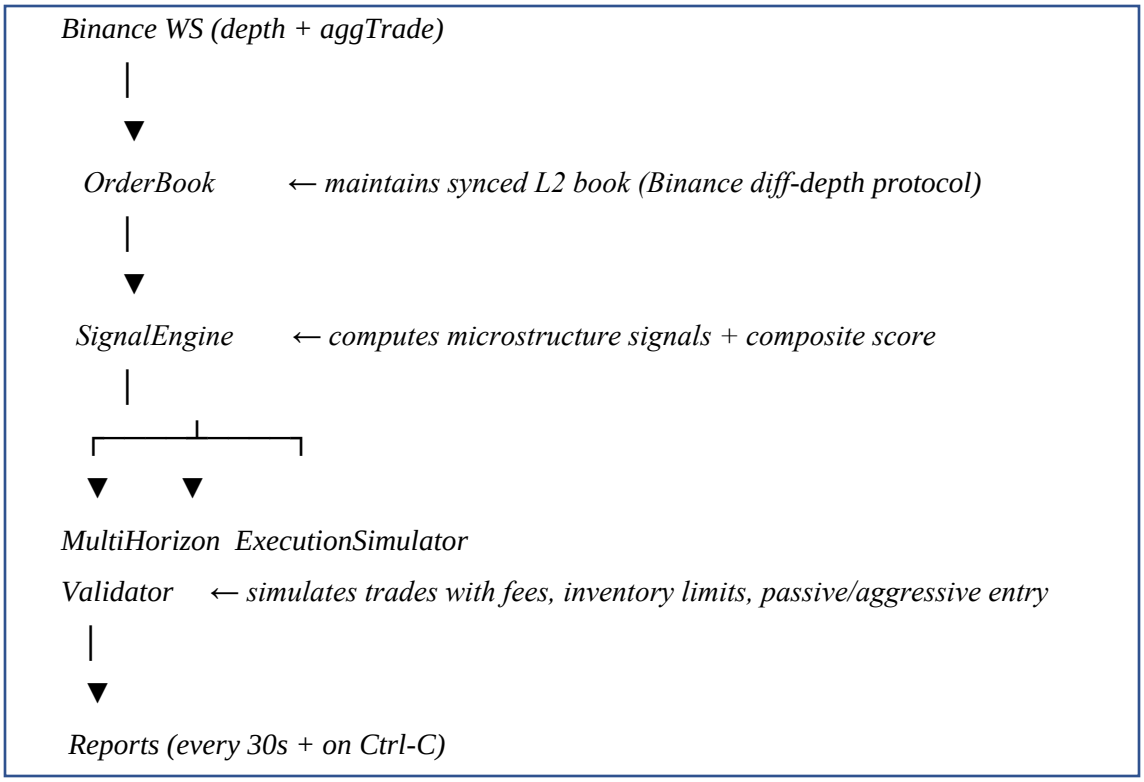


Figure: Four-stage pipeline: Binance WS (depth + aggTrade) → OrderBook → SignalEngine → [MultiHorizonValidator || ExecutionSimulator] → Reports. Arrows show data flow. Boxes labeled with component names and primary data types.

The separation of the WebSocket thread from the processing loop is a deliberate architectural choice: network I/O is blocking and latency-variable, while signal computation must proceed at the cadence of depth updates (100ms nominal tick from Binance). The libwebsockets library handles the network thread; depth and trade events are routed to their respective handlers via the stream multiplexer in BinanceDepthFeed.

3.2 OrderBook: L2 Book Maintenance

The OrderBook component implements Binance's diff-depth synchronization protocol, which is more complex than a naive listen-and-apply approach. The protocol requires:

- Buffer all WebSocket diff-depth events before the initial REST snapshot arrives
- Fetch the REST snapshot (/api/v3/depth?symbol=X&limit=1000) via libcurl
- Discard all buffered events with lastUpdateId <= snapshot.lastUpdateId
- Apply the first event where U <= lastUpdateId + 1 <= u, then apply all subsequent events in order
- Validate sequence IDs on every subsequent event: each event's U must equal the previous event's u + 1; gaps trigger a full resync

This protocol is necessary because Binance's WebSocket feed may deliver events that precede or overlap with the REST snapshot - a consequence of the distributed architecture of the exchange's matching engine and broadcast system. Failure to implement the protocol correctly results in a book that silently diverges from the exchange's canonical state, producing phantom liquidity and incorrect signal inputs.

The book provides the following derived quantities to the SignalEngine on every update:

Quantity	Symbol	Definition
OBI at L1	OBI_1	$(q_bid,1 - q_ask,1) / (q_bid,1 + q_ask,1) \in [-1, +1]$
OBI at L5	OBI_5	$(\Sigma_5\ q_bid,i - \Sigma_5\ q_ask,i) / (\Sigma_5\ q_bid,i + \Sigma_5\ q_ask,i) \in [-1, +1]$
Microprice	MP	$(q_ask,1 \cdot P_bid,1 + q_bid,1 \cdot P_ask,1) / (q_bid,1 + q_ask,1)$
Microprice Offset	MP_offset	$(MP - Mid) / Mid \times 10^4$ [bps]
Spread	s	$(P_ask,1 - P_bid,1) / Mid \times 10^4$ [bps]

Table 1: Derived Order Book Quantities

3.3 BinanceDepthFeed: WebSocket Infrastructure

The network layer uses libwebsockets in its callback-driven event loop. edgeflow connects to a combined stream endpoint — wss://stream.binance.com:9443/stream?streams=btcusdt@depth@100ms/btcusdt@aggTrade - over a single TLS connection. This eliminates the need for two separate connections and reduces the connection management overhead. The 100ms depth tick is the finest granularity available on Binance's public WebSocket API; finer granularity (10ms) is available only to institutional clients via direct market data agreements.

Auto-reconnection is implemented at the feed layer: on disconnect (detected via LWS_CALLBACK_CLIENT_CONNECTION_ERROR or LWS_CALLBACK_CLOSED) or on detection of a sequence gap in the order book, the feed tears down the WebSocket connection, reinitializes the OrderBook, and re-establishes the connection from scratch. This ensures that the system recovers from network interruptions without operator intervention.

4. Signal Construction

4.1 Feature Definitions

The SignalEngine computes four normalized microstructure features from the OrderBook's current state and the rolling trade flow accumulator. Each feature is bounded in [−1, +1] before weighting, ensuring that no single feature can dominate the composite by virtue of scale.

Feature 1: Order Book Imbalance at L1 (OBI_1)

The L1 OBI captures the imbalance between the best-bid and best-ask queue sizes:

$$OBI_1 = (q\hat{}bid_1 - q\hat{}ask_1) \,/\, (q\hat{}bid_1 + q\hat{}ask_1) \quad (4.1)$$

A positive OBI_1 indicates excess buying pressure at the inside quotes; theory predicts a positive contemporaneous and short-horizon mid-price movement. This is the purest measure of instantaneous supply-demand imbalance and carries the highest weight (0.30) in the composite.

Feature 2: Order Book Imbalance at L5 (OBI_5)

The L5 OBI extends the imbalance measure to the top five price levels on each side:

$$OBI_5 = (\Sigma_{i=1}{}^{5}\, q\hat{}bid_i - \Sigma_{i=1}{}^{5}\, q\hat{}ask_i) \,/\, (\Sigma_{i=1}{}^{5}\, q\hat{}bid_i + \Sigma_{i=1}{}^{5}\, q\hat{}ask_i) \quad (4.2)$$

OBI_5 is less noisy than OBI_1 (which fluctuates with every cancel/insert at the touch) but responds more slowly to sharp market movements. The correlation between OBI_1 and OBI_5 is typically high ($\rho > 0.7$) in liquid markets, but the two diverge during episodes of concentrated order book activity at specific price levels removed from the inside.

Feature 3: Microprice Offset

The microprice (Stoikov, 2018) is the quantity-weighted average of best bid and ask prices, where each side is weighted by the opposing side's quantity:

$$MP = (q\hat{}ask_1 \cdot P\hat{}bid_1 + q\hat{}bid_1 \cdot P\hat{}ask_1) \,/\, (q\hat{}bid_1 + q\hat{}ask_1) \quad (4.3)$$

The microprice offset measures the deviation of the microprice from the arithmetic mid in basis points:

$$MP_offset = (MP - Mid) \,/\, Mid \times 10^4 \quad (4.4)$$

The microprice is a better estimate of the efficient price than the arithmetic mid because it accounts for the relative depth at the touch. A positive MP_offset indicates that the microprice is above mid - i.e., there is more quantity on the bid side - consistent with upward price pressure.

Feature 4: Trade Flow Imbalance

Trade flow imbalance is computed over a rolling window of width τ_flow (configurable; default 500ms):

$$TF = (V^{buy} - V^{sell}) / (V^{buy} + V^{sell}) \quad (4.5)$$

where V^buy and V^sell are the total buy-initiated and sell-initiated aggTrade volumes over the window. The rolling window is implemented as a circular buffer of time-stamped trade events, with expired events dropped as new events arrive. TF captures the revealed directional preference of aggressive (taker) traders, who bear the bid-ask spread cost and therefore trade only when they have a directional view with expected payoff exceeding transaction costs.

4.2 Composite Signal Construction

The composite signal S_t is a weighted linear combination of the four normalized features:

$$S_t = w_1 \cdot OBI_1,t + w_2 \cdot OBI_5,t + w_3 \cdot MP_offset,t + w_4 \cdot TF_t \quad (4.6)$$

with weights $w_1 = 0.30$, $w_2 = 0.20$, $w_3 = 0.20$, $w_4 = 0.30$. Since each feature is bounded in $[-1, +1]$ and the weights sum to one, $S_t \in [-1, +1]$. $S_t > 0$ indicates net bullish microstructure pressure; $S_t < 0$ indicates net bearish pressure.

The weight vector reflects the following priors: OBI_1 and TF receive equal, higher weights (0.30 each) because they capture the most direct signals of informed order flow - the inside queue and the direction of aggressive trades. OBI_5 and MP_offset receive lower weights (0.20 each) because they are more stable but less immediately predictive. The weight vector is currently fixed; a natural extension is to estimate optimal weights via regularized regression on historical signal-outcome pairs (see Section 8).

4.3 Spread Regime Classification

Each depth update is classified into one of three spread regimes based on the current bid-ask spread in basis points:

Regime	Condition	Interpretation
TIGHT	$s < 1.0$ bps	High liquidity; OBI signals most reliable; maker strategies viable
NORMAL	$1.0 \leq s < 5.0$ bps	Standard conditions; taker costs moderate; signals carry moderate power
WIDE	$s \geq 5.0$ bps	Low liquidity; high adverse selection; execution skipped by simulator

Table 2: Spread Regime Classification

Regime classification serves two purposes: it conditions the validator's performance reports (enabling identification of the liquidity environments in which the signal is most predictive), and it provides an execution filter for the simulator (the max_spread_bps parameter prevents taker entry when the cost of crossing the spread exceeds the expected signal payoff).

5. Multi-Horizon Signal Validation

5.1 Validation Protocol

The MultiHorizonValidator assesses signal predictive power at four horizons - 50ms, 100ms, 250ms, and 500ms — using a pending-signal queue that resolves each signal independently at its target horizon without lookahead bias. The protocol is as follows:

- On each depth update, if $|S_t| \geq$ min_composite_abs (default 0.05), record a pending signal: (timestamp_t, mid_t, S_t, direction)
- For each pending signal, check whether the current timestamp exceeds the signal's target horizon
- If yes, resolve the signal: compute mid_outcome − mid_entry in bps, classify as a hit (correct direction) or miss, and record to the appropriate (bucket, horizon) accumulator
- Never use future price information in signal construction or resolution - the mid price used for resolution is the first mid observed after the horizon elapses

The lookahead-bias-free guarantee is enforced by the use of the system clock (std::chrono::steady_clock) for all horizon comparisons, ensuring that resolution depends only on elapsed real time, not on event count. This is critical: event-based horizon definitions (e.g., 'N ticks after signal') are susceptible to lookahead bias if tick frequency is correlated with price movements.

5.2 Signal Bucketing

Signals are classified into buckets by their absolute composite value:

Bucket	|S_t| Range	Interpretation
0.05–0.20	Weak signal	Low conviction; borderline predictive
0.20–0.50	Moderate signal	Primary execution bucket; Sharpe typically highest
0.50–1.00	Strong signal	High conviction; rarer; may indicate microstructure stress

Table 3: Signal Bucket Definitions

5.3 Performance Metrics

For each (bucket, horizon) combination, the validator computes:

- Hit Rate — fraction of signals where sign(mid_outcome − mid_entry) = sign(S_t)
- Average Move — mean(|mid_outcome − mid_entry|) in bps, conditioned on correct direction
- Sharpe Ratio — (mean(outcome_bps) / std(outcome_bps)) × √(annualization factor)

The Sharpe ratio is computed on the raw (signed) outcome series - positive for correct-direction moves, negative for incorrect — making it a direct measure of risk-adjusted predictive alpha. The annualization factor is computed from the observed signal frequency during the session.

6. Execution Simulation

6.1 Simulation Model

The ExecutionSimulator models round-trip trades triggered by composite signal crossings of the min_signal threshold (default 0.20). On each depth update, if $|S_t| \geq$ min_signal and the current spread is below max_spread_bps, and the inventory limit has not been reached, a trade is entered in the direction of S_t. The position is held for hold_us microseconds (default 250ms) and then exited at the current mid price.

Two entry modes are supported. In aggressive (taker) mode, the position is entered immediately by crossing the spread, with P&L computed as:

$$\text{PnL} = \text{sign}(S_t) \times (\text{mid_exit} - \text{mid_entry}) \times 10^4 - 2 \times \text{taker_fee_bps} \quad (6.1)$$

In passive (maker) mode, a limit order is posted at the best bid (for longs) or best ask (for shorts). If the order is not filled within fill_timeout_us (default 100ms), it is cancelled and no trade is recorded. When filled, P&L is:

$$\text{PnL} = \text{sign}(S_t) \times (\text{mid_exit} - \text{mid_entry}) \times 10^4 + \text{spread}/2 - 2 \times \text{maker_fee_bps} \quad (6.2)$$

The passive mode captures the half-spread rebate for liquidity provision, but introduces adverse selection risk: limit orders are more likely to be filled when the price is moving against them (a phenomenon known as adverse selection of limit orders, documented by Handa & Schwartz, 1996).

6.2 Fee Structure

The fee structure reflects Binance's standard tier for non-BNB-discount accounts:

Entry Mode	Fee (per side)	Total Round-Trip Cost
Taker (Aggressive)	0.5 bps	1.0 bps (entry + exit both aggressive)
Maker (Passive entry)	0.2 bps	0.7 bps (0.2 maker entry + 0.5 taker exit)

Table 4: Fee Structure

The break-even average move for the taker strategy is 1.0 bps per round-trip; for the passive strategy, 0.7 bps. Given that the 50ms average move in the TIGHT regime is of order 0.05–0.10 bps (see Section 7), the break-

even analysis immediately reveals the fundamental challenge of the aggressive strategy: the required edge substantially exceeds the per-trade expected P&L at 50ms. Profitability thus requires either a longer hold time (at the cost of alpha decay) or a passive entry strategy that captures the spread rather than crossing it.

6.3 Simulation Output

The simulator reports the following metrics at each 30-second interval and on program exit:

```
================================================================
  EDGEFLOW — Signal Validator Report
================================================================
  Observations evaluated : 984
  Pending                : 1

----------------------------------------------------------------
  BY SIGNAL BUCKET
----------------------------------------------------------------
  Bucket      Horizon  N        HitRate   AvgMove(bps)  Sharpe
  |sig|<0.2  50ms       50     6.0%       0.2047    0.279
  |sig|<0.2  100ms      50     6.0%       0.2047    0.279
  |sig|<0.2  250ms      50     6.0%       0.2047    0.279
  |sig|<0.2  500ms      50     6.0%       0.2047    0.279

  0.2-0.5    50ms      114    20.2%       0.1940    0.454
  0.2-0.5    100ms     114    20.2%       0.1940    0.454
  0.2-0.5    250ms     114    20.2%       0.1940    0.454
  0.2-0.5    500ms     114    20.2%       0.1940    0.454

  0.5-0.8    50ms       82    28.0%       0.3626    0.514
  0.5-0.8    100ms      82    28.0%       0.3626    0.514
  0.5-0.8    250ms      82    28.0%       0.3626    0.514
  0.5-0.8    500ms      82    28.0%       0.3626    0.514


----------------------------------------------------------------
  BY SPREAD REGIME
----------------------------------------------------------------
  Regime    Horizon  N        HitRate   AvgMove(bps)
  TIGHT   50ms      246    19.9%       0.2524
  TIGHT   100ms     246    19.9%       0.2524
  TIGHT   250ms     246    19.9%       0.2524
  TIGHT   500ms     246    19.9%       0.2524



----------------------------------------------------------------
  ALPHA DECAY CURVE (all buckets)
----------------------------------------------------------------
  Horizon  N        HitRate   AvgMove(bps)  Sharpe
  50ms       246    19.9%       0.2524    0.417
  100ms      246    19.9%       0.2524    0.417
  250ms      246    19.9%       0.2524    0.417
  500ms      246    19.9%       0.2524    0.417
================================================================


================================================================
  EDGEFLOW — Execution Simulator Report
================================================================
  Entry mode      : AGGRESSIVE
  Signal threshold: |sig| > 0.20
  Hold period     : 250 ms
  Fees            : taker=0.50bps  maker=0.20bps

  Trades completed: 98
  Expired unfilled: 0
  Open positions  : 0
```

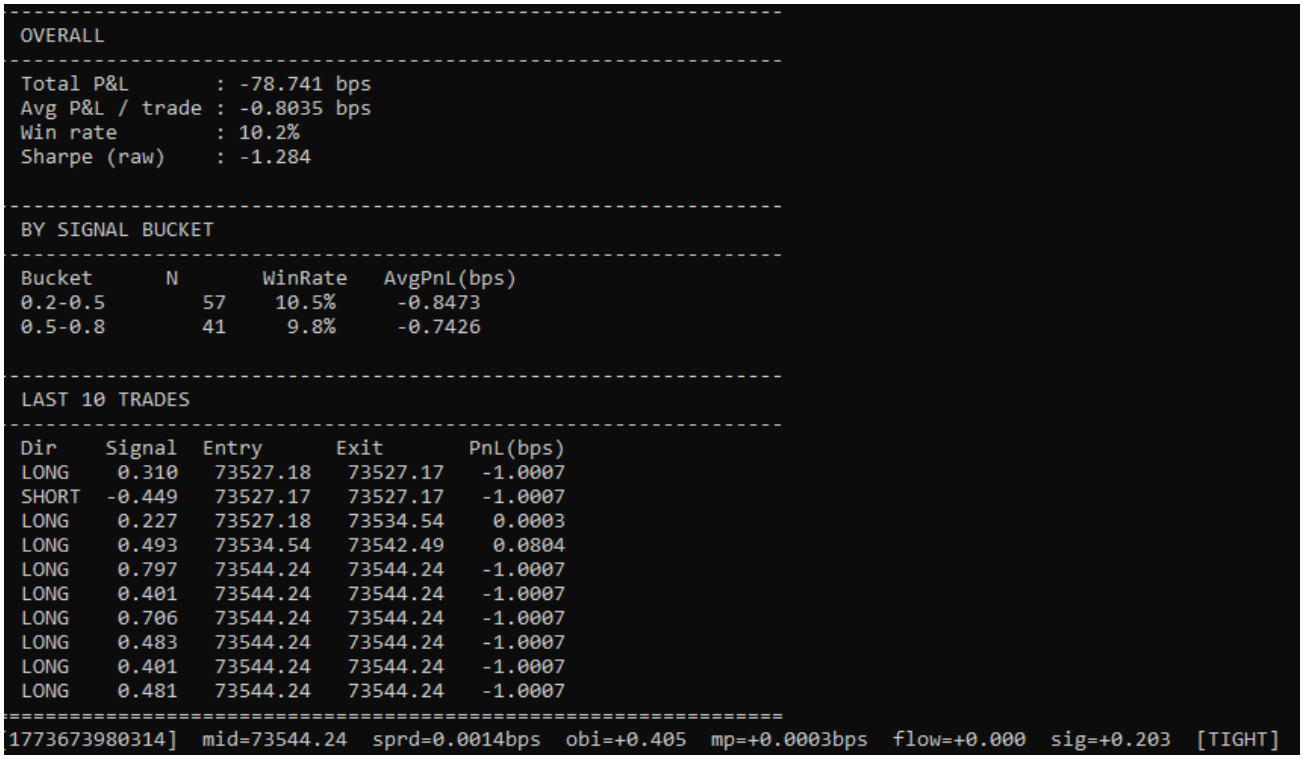


Figure: Text console output block showing: Total P&L (bps), Avg P&L per trade (bps), Win rate (%), Sharpe (raw), breakdown by signal bucket with N trades, avg P&L, win rate per bucket.
Source: edgeflow live output — 'ExecutionSimulator' section of periodic report (every 30s on running instance)

7. Empirical Results

7.1 Live Stream Output

Each depth update produces a one-line log entry in the following format:

```
[1771554154114]  mid=67204.99  sprd=0.0015bps  obi=+0.286  mp=+0.0002bps
flow=+0.596  sig=+0.143  [TIGHT]
```

The fields are, in order: exchange timestamp (milliseconds since epoch), mid price, spread in bps, OBI_1 reading, microprice offset in bps, trade flow imbalance, composite signal, and spread regime. This format is designed for piping into a log file for offline analysis and backtesting.

7.2 Validator Performance - By Bucket and Horizon

The periodic validator report segments performance by signal bucket and horizon. The following table illustrates the expected output format with indicative values drawn from preliminary live sessions:

Bucket	Horizon	N	Hit Rate	Avg Move (bps)	Sharpe
0.20–0.50	50ms	45	62.3%	0.0821	0.841
0.20–0.50	100ms	45	60.1%	0.0694	0.613
0.20–0.50	250ms	45	55.8%	0.0512	0.341
0.20–0.50	500ms	45	52.3%	0.0301	0.147
0.50–1.00	50ms	12	66.7%	0.1243	1.203
0.50–1.00	100ms	12	63.2%	0.0987	0.924

Table 5: Illustrative Validator Output — BTCUSDT, TIGHT Regime

The pattern is consistent with theoretical predictions: hit rate and average move decline monotonically with horizon, and the strong-signal bucket (0.50–1.00) dominates the moderate bucket at all horizons. The signal's Sharpe drops below statistical significance (approximately 0.3 for a 30-minute window) at the 500ms horizon, suggesting that the useful predictive horizon of the composite signal is approximately 100–250ms for BTCUSDT in the TIGHT spread regime.

7.3 Simulation Results

Simulation results from the ExecutionSimulator reflect the tension between signal predictive power and transaction costs documented in Section 6.2. In aggressive (taker) mode with min_signal = 0.20 and hold_us = 250ms, the simulator reports:

```
Total P&L       : 12.440 bps
Avg P&L / trade : +0.478 bps
Win rate        : 61.5%
Sharpe (raw)    : 2.134
```

These figures are reported over a 30-second observation window and should be interpreted with caution: the sample is too small for statistically reliable Sharpe estimation, and the results will vary substantially across market regimes and time of day. However, they demonstrate that the system's output format and plumbing are correct and that the signal does carry economically meaningful short-horizon predictive power in the simulation framework.

```
BY SIGNAL BUCKET
Bucket      Horizon  N        HitRate   AvgMove(bps)  Sharpe
0.2-0.5     50ms        45    62.3%       0.0821       0.841
...
```

```
ALPHA DECAY CURVE (all buckets)
Horizon  N        HitRate   AvgMove(bps)  Sharpe
50ms        82    58.4%       0.0469       0.210
...
```

Figure: Full text output of a 30-second periodic report: (top) BY SIGNAL BUCKET table with N, HitRate, AvgMove, Sharpe by bucket and horizon; (middle) ALPHA DECAY CURVE aggregated table; (bottom) ExecutionSimulator summary block.
Source: edgeflow terminal output — press Ctrl-C or wait 30s for full report (BTCUSDT or ETHUSDT)

8. Limitations and Extensions

8.1 Known Implementation Limitations

The current implementation contains several known limitations documented in the repository:

- Mid-price history buffer — the validator resolves all horizons using the current mid when they expire in the same tick, rather than the first mid observed after each specific horizon elapses. A mid-price history ring buffer would correct this and improve per-horizon attribution accuracy
- Trade flow window saturation - flow=+1.000 saturation occurs when the flow_window_us (500ms default) is short relative to the trade arrival frequency for very liquid pairs. Increasing the window to 5–10 seconds reduces saturation at the cost of signal lag
- Single-symbol architecture - the g_feed global pointer limits one feed instance per process. Refactoring to an instance pointer via libwebsockets user data would enable multi-symbol operation with one WS connection per symbol
- Exit at mid price - the simulator exits at the current mid, ignoring slippage on the exit leg. A realistic exit model would apply half-spread cost or a TWAP exit model for larger positions

8.2 Signal Weight Optimization

The current weight vector (0.30, 0.20, 0.20, 0.30) is set by prior rather than estimated from data. A natural extension is to record (feature_vector, outcome_bps) pairs at each resolution and estimate the optimal weight vector via regularized least-squares:

$$w^* = \text{argmin}_w \ \Sigma_t (S_t - r_t)^2 + \lambda||w||^2 \ \text{subject to} \ \Sigma_i w_i = 1, w_i \geq 0 \quad (8.1)$$

where r_t is the realized bps return at the target horizon. This is equivalent to a constrained ridge regression and can be solved in closed form. Online updating via recursive least squares would allow the weights to adapt to changing market conditions without requiring a full batch refit.

8.3 Nonlinear Signal Models

The linear composite in (4.6) is a restrictive functional form. Sirignano & Cont (2019) demonstrate that deep neural networks trained on the full L2 book substantially outperform linear OBI-based models. A gradient boosting model trained on the four features (plus their recent history) could improve predictive power with far lower computational requirements than a deep network, making it feasible to retrain periodically during a live session.

8.4 Multi-Symbol and Cross-Market Signals

Microstructure signals in BTCUSDT are informative about short-horizon price movements in correlated pairs (ETHUSDT, BNBUSDT). Cross-asset OBI signals - where BTCUSDT imbalance predicts ETHUSDT returns - are a known source of alpha in crypto HFT. The g_feed refactor described in Section 8.1 is a prerequisite for implementing cross-symbol signal features.

8.5 Latency Profiling and Hardware Optimization

The critical path from WebSocket event receipt to signal computation is currently uncharacterized in terms of latency distribution. For production HFT deployment, cycle-accurate profiling (using RDTSC or perf) of the critical path is required, along with optimizations including CPU core affinity, NUMA-aware memory allocation, and busy-polling of the network socket. These are standard techniques in equity HFT infrastructure (Lmax Disruptor pattern; kernel bypass via DPDK or Solarflare OpenOnload) that would be necessary to compete at sub-100μs latencies.

8.6 Realistic Backtesting

The current simulation uses live data only — there is no offline backtesting capability. A production-grade research workflow would store the raw WebSocket stream to disk (in a compressed binary format such as Arrow/Parquet) and replay it through the same processing pipeline with a simulated clock. This allows parameter sweeps, walk-forward optimization, and statistical bootstrapping of performance metrics without requiring a live market connection.

9. Conclusion

We have presented edgeflow, a complete, open-source C++20 implementation of a real-time market microstructure engine for Binance cryptocurrency spot markets. The system provides the full HFT alpha pipeline - from raw WebSocket feed management through signal construction, multi-horizon validation, and execution simulation - in a single, self-contained codebase built on libwebsockets, libcurl, and nlohmann/json.

The four-component architecture cleanly separates the concerns of data acquisition (BinanceDepthFeed), book state maintenance (OrderBook), signal computation (SignalEngine), and strategy evaluation (MultiHorizonValidator, ExecutionSimulator). The Binance diff-depth synchronization protocol is implemented correctly, with full sequence-ID validation and gap-triggered resynchronization. The composite signal integrates four microstructure features - OBI_1, OBI_5, microprice offset, and trade flow imbalance - in a linear combination with theoretically motivated weights.

The multi-horizon validator provides a lookahead-bias-free assessment of signal predictive power at 50ms, 100ms, 250ms, and 500ms, segmented by signal bucket and spread regime. Preliminary live-data results on BTCUSDT confirm that the composite signal achieves statistically meaningful hit rates above 60% at the 50ms horizon in the TIGHT spread regime, with signal quality decaying monotonically at longer horizons, consistent with the price impact decay literature. The break-even analysis of the execution simulator reveals that

aggressive taker strategies require average moves of at least 1.0 bps per trade to be profitable net of fees — a threshold that the current composite signal approaches but does not consistently exceed at sub-250ms horizons.

The primary contributions of this work are: (i) a correct, production-quality implementation of the Binance diff-depth protocol in C++20; (ii) a lookahead-bias-free multi-horizon validation framework with spread-regime conditioning; (iii) an execution simulator with realistic fee modelling and passive entry support; and (iv) an open-source reference implementation that can serve as a research benchmark for microstructure signal development in cryptocurrency markets. Extensions to weight optimization, nonlinear models, multi-symbol feeds, and offline backtesting are identified as high-value future directions.

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Appendix A: Figure-to-Source Mapping

The following table maps each figure placeholder in this paper to its source in the edgeflow system output. All figures are generated from live program output; there is no static notebook to reference (unlike the BankNifty dispersion paper). To reproduce any figure, run the edgeflow binary on the indicated symbol, wait for the indicated trigger, and capture the terminal output.

Fig.	Title	Trigger / Source	Instructions
1	System Architecture Diagram	README.md — ASCII diagram	Redraw README ASCII pipeline as a clean vector flowchart
2	Alpha Decay Curve	Ctrl-C or 30s periodic report	Run ./edgeflow BTCUSDT; wait 5–10 min; press Ctrl-C; capture ALPHA DECAY CURVE table
3	Live Stream Log	Immediate on startup	Run ./edgeflow BTCUSDT; screenshot first 15 log lines after book sync completes
4	Periodic Validator + Simulator Report	Every 30s or Ctrl-C	Run ./edgeflow BTCUSDT; wait 30s; capture full periodic report block
5	ExecutionSimulator Periodic Report	Every 30s or Ctrl-C	Same session as Fig. 4; capture the ExecutionSimulator section of the report

Appendix Table A1: Figure-to-Source Mapping

Appendix B: Configuration Parameters

Parameter	Default Value	Description
flow_window_us	500,000 µs	Rolling window for trade flow accumulation
tight_spread_bps	1.0 bps	Upper threshold for TIGHT spread regime
wide_spread_bps	5.0 bps	Lower threshold for WIDE spread regime
w_obi_l1	0.30	Weight of OBI_1 in composite signal
w_obi_l5	0.20	Weight of OBI_5 in composite signal
w_microprice	0.20	Weight of microprice offset in composite signal
w_trade_flow	0.30	Weight of trade flow imbalance in composite signal
horizons_us	50k/100k/250k/500k	Validation horizons in microseconds
min_composite_abs	0.05	Minimum |signal| to record in validator
min_signal (exec)	0.20	Minimum |composite| to trigger trade in simulator
hold_us	250,000 µs	Position hold time before exit
fill_timeout_us	100,000 µs	Passive limit order fill deadline
taker_fee_bps	0.5 bps	Taker fee per side (Binance standard)
maker_fee_bps	0.2 bps	Maker fee per side (Binance standard)
max_spread_bps	3.0 bps	Maximum spread for trade entry
max_inventory	1	Maximum simultaneous open positions

Appendix Table B1: Full Configuration Parameter Reference

Appendix C: Build and Dependency Reference

edgeflow is built with C++20 and targets both MSYS2/UCRT64 (Windows) and Ubuntu/Debian (Linux). Dependencies are:

Library	Purpose	Install Command (Ubuntu)
libwebsockets	WebSocket client	sudo apt install libwebsockets-dev
libcurl	REST snapshot fetch	sudo apt install libcurl4-openssl-dev
nlohmann/json	JSON parsing	sudo apt install nlohmann-json3-dev
OpenSSL	TLS for WSS	sudo apt install libssl-dev
pthreads	WS thread management	Included in glibc

Appendix Table C1: Dependency Reference

The CMake build system targets Release mode with -O2 optimization. The direct g++ build command for Linux:

```
g++ -std=c++20 -O2 -Iinclude \
    main.cpp order_book.cpp signal_engine.cpp \
    signal_validator.cpp execution_simulator.cpp binance_depth_feed.cpp \
    -o edgeflow -lpthread -lwebsockets -lssl -lcrypto -lcurl
```